

Full Marks: 30

Time: 2 Hours

Answer Question-1 and any three from the remaining.

1. (a) What advantages are there to a language-processing system in which the compiler produces assembly language rather than machine language?
(b) Explain tokens, patterns, and lexemes. Demonstrate the same with examples.

[3 + 3 = 6]

2. (a) Write regular expressions for specifying identifiers and constants of C. Discuss how finite automata is used to represent tokens and performs lexical analysis with examples..
(b) Consider the following Context Free Grammar where S is the start symbol:
 $S \rightarrow SS + | SS * | a$. and the string $aa + a*$.
i. Give the leftmost derivation of the string.
ii. Give rightmost derivation of the string.
iii. Give the parse tree of the string.
iv. Is the grammar ambiguous or unambiguous? Justify your answer.

[4 + 4 = 8]

3. Consider the following Grammar, $G = (\{A, B\}, \{a, b, c, d\}, P, A)$ where P is set of production rules as follows:

$A \rightarrow Aa | Aab | Bc$,

$B \rightarrow BAa | d$

- (a) Eliminate left recursion from the above grammar.
(b) Compute FIRST & FOLLOW set for the non-terminals. Check the grammar is LL(1) or not; Show the parsing for $dcab$.

[2 + 6 = 8]

4. (a) What are the use of shift reduces parser? Explain conflicts that may occur during shift-reduce parsing.
(b) What do you mean by handle pruning in bottom-up parsing? Explain with the help of the grammar $S \rightarrow SS + | SS * | a$ and input string $aaa * a + +$. In each reduction indicate the corresponding handle.

[3 + 5 = 8]

5. Check whether the following grammar is SLR (1) or not. Explain your answer with Reasons.

$S \rightarrow L = R$

$S \rightarrow R$

$L \rightarrow *R$

$L \rightarrow id$

$R \rightarrow L$

[5 + 3 = 8]

Indian Institute of Engineering Science and Technology, Shibpur
Dual Degree (B.Tech. - M.Tech.) 7th Semester (CST) Examination, November, 2025
Compiler Design (CS 4101)

Full Marks: 50

Time: 3 Hours

Answer Question-1 and any four from the remaining.

1. (a) Explain the error handling and error recovery mechanism of syntax analyser .
(b) Explain the following peephole optimization techniques: a) Elimination of Redundant Code, b) Elimination of Unreachable Code.

[4 + (1 + 1) = 6]

2. (a) Consider the following Context Free Grammar where S is the start symbol:

$S \rightarrow L = R$

$S \rightarrow R$

$L \rightarrow *R$

$L \rightarrow id$

$R \rightarrow L$

Check Whether the grammar is ambiguous or not.

- (b) What is recursive descent parsing? List the problems faced in designing such a parser.
(c) Construct a Finite Automata equivalent to the regular expression.

$(0 + 1)^*(00 + 11)(0 + 1)^*$

[4 + 4 + 3 = 11]

3. Consider the following grammar production rules where S is the start symbol:

$S \rightarrow ABD$

$A \rightarrow a|DB|\epsilon$

$B \rightarrow gD|dA|\epsilon$

$D \rightarrow e|f$

- (a) Construct FIRST and FOLLOW for each non terminal of the above grammar.
(b) Construct the predictive parsing table for the above grammar.
(c) Show the parsing on a valid string and on a invalid string.
(d) Check whether the grammar is LL(1). Give justification.

[3 + 3 + 3 + 2 = 11]

4. (a) Explain the model of shift reduces parser? Explain conflicts that may occur during shift-reduce parsing.

- (b) Construct the CLR parsing table for the following grammar where S is the start symbol:

$S \rightarrow CC$

$C \rightarrow cC|d|e$

[5 + 6 = 11]

5. (a) Write down the algorithm to find the leader in basic block. Write down the three-address code and construct the basic blocks for the following program segment.

```
sum = 0; i = 0;
while (i <= 10)
{
    sum = sum + a[i];
    i++;
}
```

where the datatype for a, b and x are integer.

- (b) Why symbol-table is needed in various phases of compilers? How hashing can be used to design symbol-table.
- (c) Explain the characteristics of peephole code optimization technique.

[3 + 5 + 3 = 11]

6. (a) Explain the simplification of simple type checker for statements, expressions and functions.
- (b) Explain Loop optimization in detail using suitable example.

[6 + 5 = 11]

7. (a) Explain the sequence of the stack allocation process for a function call using a suitable example.
- (b) Translate the expression $-(a+b)*(c+d)+(a+b+c)$ into: (i) quadruples, (ii) triples and (iii) indirect triples.
- (c) Define the activation record. What are the contents of activation record?

[3 + 5 + 3 = 11]

Full Marks: 30

Time: 2 Hours

Answer Question-1 and any three from the remaining.

1. (a) List out the functions of a Lexical Analyzer. State the reasons for the separation of Analysis programs into Lexical, Syntax, and Semantic Analyses.
(b) Explain the various errors encountered in different phases of compiler.

[3 + 3 = 6]

2. (a) Write regular expressions for specifying identifiers and constants of C. Discuss how finite automata is used to represent tokens and performs lexical analysis with examples..
(b) Explain panic mode error recovery strategy for predictive parsing method using a suitable example.

[4 + 4 = 8]

3. Consider the following Grammar production rules where E is the start symbol:

$$E \rightarrow E + T | T$$

$$T \rightarrow TF | F$$

$$F \rightarrow F * | a | b$$

- (a) Eliminate left recursion from the above grammar.
(b) Compute FIRST & FOLLOW set for the non-terminals. Check the grammar is LL(1) or not; Show the parsing for $a + a + a$.

[2 + 6 = 8]

4. (a) What are the use of shift reduces parser? Explain conflicts that may occur during shift-reduce parsing.
(b) What do you mean by handle pruning in bottom-up parsing? Explain with the help of the grammar $S \rightarrow SS + | SS * | a$ and input string $aaa * a + +$. In each reduction indicate the corresponding handle.

[3 + 5 = 8]

5. What is LALR(1) grammar? Construct LALR parsing table for the following grammar:

$$S \rightarrow CC$$

$$C \rightarrow cC$$

$$C \rightarrow c | d$$

[3 + 5 = 8]

Indian Institute of Engineering Science and Technology, Shibpur
Dual Degree (B.Tech. - M.Tech.) 7th Semester (CST) Examination, November, 2024
Compiler Design (CS 4101)

Full Marks: 50

Time: 3 Hours

Answer Question-1 and any four from the remaining.

1. (a) Describe hash-table based data structures for symbol table management.
(b) Write the regular expressions to describe languages consisting of strings made of even numbers of a and b .

[3 + 3 = 6]

2. (a) What is meant by lexical analysis? Identify the lexemes that make up the token in the following program segment. Indicate the corresponding token and pattern.

```
void swap(int i, int j)
{
    int t;
    t = i;
    i = j;
    j = t;
}
```

- (b) Draw the transition diagram for relational operators and unsigned numbers.

[4 + (4 + 3) = 11]

3. Consider the following grammar production rules where S is the start symbol:

$E \rightarrow E + T | T$

$T \rightarrow TF | F$

$F \rightarrow F * | a | b$

- (a) Write the rules to eliminate left recursion in a grammar. Eliminate left recursion from the above grammar.
- (b) Compute FIRST & FOLLOW set for the non-terminals. Check the grammar is LL(1) or not; Show the parsing for $a + b + a$.

[3 + (4 + 1 + 3) = 11]

4. (a) Explain the model of shift reduces parser? Explain conflicts that may occur during shift-reduce parsing.
- (b) Construct the CLR parsing table for the following grammar where S is the start symbol:

$S \rightarrow CC$

$C \rightarrow cC | d | e$

[5 + 6 = 11]

5. (a) Describe the evaluation order of Syntax Directed Translation(SDT) with an example.
 (b) Generate intermediate code for the following code segment along with the required syntax directed definition:

```

    if (a > b)
        x = a+b;
    else
        x = a-b;
  
```

Here datatype for x, a and b are int.

[6 + 5 = 11]

6. (a) Discuss the following: i) Dead code elimination and ii) copy propagation.
 (b) Construct the DAG for the following sequence of codes:

```

1. t1:=4*i
2. t2:=a[t1]
3. t3:=4*i
4. t4:=b[t3]
5. t5:=t2*t4
6. t6:=prod+t5
7. prod:=t6
8. t7:=i+1
9. i:=t7
10. if i<=20 goto 1
  
```

- (c) Explain loop optimization in detail using a suitable example.

[4 + 3 + 4 = 11]

7. (a) Explain the sequence of the stack allocation process for a function call using a suitable example.
 (b) Translate the expression $-(a+b)*(c+d)+(a+b+c)$ into: (i) quadruples, (ii)triples and (iii) indirect triples.
 (c) Define the activation record. What are the contents of activation record?

[3 + 5 + 3 = 11]

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. - M.Tech. Dual Degree 7th Semester (CST) Examination (Mid Semester) 2023
Compiler Design (CS 4101)

Full Marks: 30

Time: 2 Hours

Answer Question-1 and any three from the remaining.

Do all parts of a question together. Do not mix up answers to parts of different questions in the answer script.

1. (a) List out the functions of a Lexical Analyzer. State the reasons for the separation of Analysis programs into Lexical, Syntax, and Semantic Analyses.
(b) Explain the various errors encountered in different phases of compiler.

[3 + 3 = 6]

2. (a) Discuss how finite automata is used to represent tokens and performs lexical analysis with examples.
(b) Explain panic mode error recovery strategy for predictive parsing method using a suitable example.

[3 + 5 = 8]

3. Consider the following Grammar production rules where E is the start symbol:

$$E \rightarrow E + T | T$$

$$T \rightarrow TF | F$$

$$F \rightarrow F * | a | b$$

- (a) Eliminate left recursion from the above grammar.
(b) Compute FIRST & FOLLOW set for the non-terminals. Check the grammar is LL(1) or not; Show the parsing for $a + a + a$.

[2 + 6 = 8]

4. (a) What are the use of shift reduces parser? Explain conflicts that may occur during shift-reduce parsing.
(b) What do you mean by handle pruning in bottom-up parsing? Explain with the help of the grammar $S \rightarrow SS + | SS * | a$ and input string $aaa * a + +$. In each reduction indicate the corresponding handle.

[3 + 5 = 8]

5. Construct the SLR sets of items for the grammar where E is the start symbol:

$$E \rightarrow E + T | T : T \rightarrow TF | F : F \rightarrow F * | a | b$$

Show the SLR parsing table for this grammar. Is the grammar SLR?

[5 + 3 = 8]

Indian Institute of Engineering Science and Technology, Shibpur
Dual Degree (B.Tech. - M.Tech.) 7th Semester (CST) Examination (End Semester)
November, 2023
Compiler Design (CS 4101)

Full Marks: 50

Time: 3 Hours

Answer Question-1 and any four from the remaining.

Do all parts of a question together. Do not mix up answers to parts of different questions in the answer script.

1. (a) Write regular expressions to specify the identifiers and constants of C.
(b) What do you mean by left factoring of grammar? Explain.
(c) What is a handle in bottom up parsing? Explain.

[2 + 2 + 2 = 6]

2. Consider the following Grammar production rules where S is the start symbol:

$S \rightarrow ACB|CbB|Ba; A \rightarrow da|BC; B \rightarrow g|\epsilon; C \rightarrow h|\epsilon$

- (a) Eliminate left recursion from the above grammar.
(b) Compute FIRST & FOLLOW set for the non-terminals. Check the grammar is LL(1) or not; Show the parsing for "ghhg".

[3 + 4 + 1 + 3 = 11]

3. (a) Explain the model of shift reduces parser? Explain conflicts that may occur during shift-reduce parsing.
(b) Construct the CLR parsing table for the following grammar where S is the start symbol:

$S \rightarrow L = R|R$

$L \rightarrow *R|id$

$R \rightarrow L$

[5 + 6 = 11]

4. (a) Explain the use of symbol table in compilation process. List out the various attributes for implementing the symbol table.
(b) Generate intermediate code for the following code segment along with the required syntax directed definition:

```
if (a > b)
    x = a+b;
else
    x = a-b;
```

Here datatype for x, a and b are int.

[6 + 5 = 11]

5. (a) Explain the algebraic translations of local machine-independent optimizations.
(b) Discuss the following: i) Dead code elimination and ii) copy propagation.
(c) Explain loop optimization in detail using a suitable example.

[3 + 4 + 4 = 11]

6. (a) Translate the expression $-(a+b)*(c+d)+(a+b+c)$ into: (i) quadruples, (ii)triples and (iii) indirect triples.
(b) List the field in an activation record. Write down the purpose of each of these fields in an activation record.
(c) Explain the sequence of stack allocation process for a function call using suitable example.

[3 + 5 + 3 = 11]

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. - M.Tech. Dual Degree 7th Semester (CST) Examination (Mid Semester) 2022
Compiler Design (CS 4101)

Full Marks: 30

Time: 2 Hours

Answer Question-1 and any three from the remaining.

Do all parts of a question together. Do not mix up answers to parts of different questions in the answer script.

1. (a) Discuss the role of lexical analyzer in detail with necessary examples.
(b) Explain the various errors encountered in different phases of compiler.

[3 + 3 = 6]

2. (a) Discuss how finite automata is used to represent tokens and performs lexical analysis with examples.
(b) Explain panic mode error recovery strategy for predictive parsing method using a suitable example.

[3 + 5 = 8]

3. (a) What do you mean by handle pruning in bottom-up parsing? Explain with the help of the grammar $S \rightarrow SS + | SS * | a$ and input string $aaa * a + +$. In each reduction indicate the corresponding handle.
(b) Explain the conflicts that may occur during shift reduce parsing.

[5 + 3 = 8]

4. Consider the following Grammar production rules where S is the start symbol:

$S \rightarrow ACB | CbB | Ba; A \rightarrow da | BC; B \rightarrow g | \epsilon; C \rightarrow h | \epsilon$

- (a) Eliminate left recursion from the above grammar.
- (b) Compute FIRST & FOLLOW set for the non-terminals. Check the grammar is LL(1) or not; Show the parsing for "ghhg".

[2 + 6 = 8]

5. Construct the SLR sets of items for the grammar where E is the start symbol:

$E \rightarrow E + T | T; T \rightarrow TF | F; F \rightarrow F * | a | b$

Show the SLR parsing table for this grammar. Is the grammar SLR?

[5 + 3 = 8]

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. - M.Tech. Dual Degree 7th Semester (CST) Examination, 2022
Compiler Design (CS 4101)

Full Marks: 50

Time: 3 Hours

*Answer Question-1 and any four from the remaining.
Do all parts of a question together. Do not mix up answers to parts of different questions in
the answer script.*

1. (a) What are compiler construction tools? Write note on each compiler Construction tool.
(b) Eliminate Left recursion from the grammar $A \rightarrow Ac|Aad|bd$

[3 + 3 = 6]

2. Consider the following following grammar

$S \rightarrow CC$

$C \rightarrow cC|d$

where S is the start symbol, set of terminals are c, d .

- (a) Construct the sets of LR(1) items.
(b) Construct the LALR parsing table for this grammar.
(c) Show the parsing actions using the parsing table generated in (b) on the in put "cdd".

[5 + 3 + 3 = 11]

3. (a) Explain the simplification of simple type checker for statements, expressions and functions.
(b) Explain Loop optimization in detail using suitable example.

[6 + 5 = 11]

4. (a) Write down the algorithm to find the leader in basic block. Write down the three-address code and construct the basic blocks for the following program segment.

```
sum = 0; i = 0;
while (i <=10)
{
    sum = sum + a[i];
    i++;
}
```

where the datatype for a, b and x are integer.

- (b) Why symbol-table is needed in various phases of compilers? How hashing can be used to design symbol-table.
(c) Explain the characteristics of peephole code optimization technique.

[3 + 5 + 3 = 11]

5. (a) List the various error recovery strategies for a lexical analysis.
(b) Generate Intermediate code for the following code segment along with the required syntax directed translation scheme:

```
if (a > b)
    x = a + b;
else
    x = a - b;
```

where the datatype for a, b and x are integer.

- (c) Discuss how register allocation and evaluation order plays an important role in a target code generation?

$$[3 + 5 + 3 = 11]$$

6. (a) Translate the expression $-(a+b)*(c+d)+(a+b+c)$ into: (i) quadruples, (ii) triples and (iii) indirect triples.
- (b) List the field in an activation record. Write down the purpose of each of these fields in an activation record.
- (c) Explain the sequence of stack allocation process for a function call using suitable example.

$$[3 + 5 + 3 = 11]$$

28/11/19

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
B.Tech-M.Tech Dual Degree (CST) 7th Semester Examination, 2019
Compiler Design (CS - 701)

Time: 3 hours

Full Marks: 70

(Answer Question No 1 and any five from the remaining)

1. (a) What is phrase level error recovery?
(b) What is code motion?
(c) Why is it necessary to generate intermediate code instead of generating target program directly?
(d) What are the various methods of implementing three address statements?
(e) What are the properties of optimizing compiler?

[2 + 2 + 2 + 2 + 2 = 10]

2. (a) Write regular definition for the following language over {0,1} a string of 0's and 1's, which has 0 at the third position when counted from right.
(b) Draw the transition diagram for the lexical analyzer that recognize the tokens like identifiers and relational operators. Use the following rules to form the identifier:
 - begins with an alphabet
 - consists of alphabets, digits and hyphen
 - should not end with an hyphen
 - not two hyphens appear together
 (c) What is a symbol table? List the functionalities of a symbol table.

[3 + (3 + 3) + 3 = 12]

3. Consider the following Grammar

$$\begin{aligned} S &\rightarrow X|ay \\ X &\rightarrow xXy|Y \\ Y &\rightarrow a \end{aligned}$$

where S is the start symbol, set of non-terminals are $\{a, x, y\}$ and set of terminals are $\{5, 3, a, b\}$

- (a) Compute the set of SLR(1) items for this grammar
- (b) Is this grammar SLR(1)? Briefly explain why or why not.

[8 + 4 = 12]

4. (a) Write an attribute grammar to count the number of digits to the right of a decimal point or a floating point value of a decimal number given by the following grammar: (Hint: Use a *count* attribute to count the number of digits to the right of the decimal point.)

$$\begin{aligned} dnum &\rightarrow num.num \\ num &\rightarrow num\ digit|digit \\ digit &\rightarrow 0|1|2|3|4|5|6|7|8|9 \end{aligned}$$

- (b) Draw a flow diagram for the *FOR* loop and subsequently propose the semantic rules to generate three address code for the same.
- (c) Write the rules to partition a sequence of 3 address statements for forming basic blocks.

[4 + 4 + 4 = 12]

5. Consider the following grammar G for arithmetic expressions:

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T * F \mid F \\ F &\rightarrow (E) \mid id \end{aligned}$$

where E is the start symbol, set of non-terminals are $\{E, T, F\}$ and set of terminals are $\{+, (,), id\}$.

- Compute *FIRST* and *FOLLOW* for each nonterminal in G .
- Construct a predictive parsing table for G .
- Show how your predictive parser processes the input string:

$$id + id * id$$

Draw the parse tree traced out by your parser.

$$[6 + 3 + 3 = 12]$$

6. (a) List the fields in an activation record. Write down the purpose of each of these fields in an activation record.
- (b) Explain the sequence of stack allocation process for a function call using a suitable example.
- (c) Why symbol-table is needed in various phases of compilers? How hashing can be used to design symbol-table.

$$[4 + 4 + 4 = 12]$$

7. (a) What are the different types of errors a program can contain? List out different error handling strategies.
- (b) Explain the following code optimization techniques with examples: Constant propagation, Strength reduction and Code Motion.
- (c) Discuss how induction variables can be detected and eliminated from the given intermediate code:

```
B2: i= i+1
    t1=4*j
    t2=a[t1]
    if t2<10 goto B2
```

$$[3 + 6 + 3 = 12]$$

5/12/18

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
B.Tech-M.Tech Dual Degree (CST) 7th Semester Examination, 2018
Compiler Design (CS - 701)

Time: 3 hours

Full Marks: 70

(Answer Question No 1 and any five from the remaining)

1. (a) What is code motion?
(b) What do you mean by machine dependent and machine independent optimization?
(c) List the different storage allocation strategies.
(d) Convert the expression $a = b * -c + b * -c$ into three address statements.
(e) Why is it necessary to generate intermediate code instead of generating target program directly?

[2 + 2 + 2 + 2 + 2 = 10]

2. (a) Give a regular definition for e-mail addresses. An address consists of a name, followed by the separator @, and a domain. A domain is a sequence of at least two names, which are separated by a dot. Names are non-empty sequences of lowercase letters, digits and underscores (but underscores are not allowed in a leading position). For instance, `first_family@here.there` is valid, but `name@domain` (missing dot in domain) and `_name@ac._de` (leading underscores) are not valid.
(b) What is a symbol table? List the functionalities of a symbol table. Describe any two methods of implementing a symbol table.

[3 + 9 = 12]

3. Consider the following Grammar

$$\begin{aligned} E &\rightarrow 5 + T \mid 3 - T \\ T &\rightarrow V \mid V * V \mid V + V \\ V &\rightarrow a \mid b \end{aligned}$$

where E is the start symbol, set of non-terminals are {E, T, V} and set of terminals are {5, 3, a, b}

- (a) Do the left factoring for the above grammar and compute the *FIRST* and *FOLLOW* set.
(b) Construct a Predictive parsing table and check if the grammar is LL(1).
(c) Show the parsing for the sentence, $5 + a * b$.

[6 + 3 + 3 = 12]

4. Consider the following grammar

$$\begin{aligned} S &\rightarrow CC \\ C &\rightarrow cC \mid d \end{aligned}$$

where S is the start symbol, set of terminals are {c, d}

- (a) Construct the sets of LR(1) items.
(b) Construct the LALR parsing table for this grammar.
(c) Show the parsing actions using the parsing table generated in (b) on the input "cdd".

[5 + 4 + 3 = 12]

5. (a) What do you mean by attributed grammars? Discuss the translation scheme for converting an infix expression to its equivalent postfix form.
(b) Draw a flow diagram for the *FOR* loop and subsequently propose the semantic rules to generate three address code for the same.
(c) Write the rules to partition a sequence of 3 address statements for forming basic blocks.

[4 + 4 + 4 = 12]

6. (a) What is activation record? Explain the task of caller and callee when a procedure is called and returned from the procedure?
- (b) Consider the following program:

1. int foo(int a){ 2. int b = a++; 3. return b*a; 4. }	5. int bar(int b){ 6. return foo(b); 7. }	8. int main(){ 9. int c; 10. c=6; 11. return bar(c); 12. }
---	---	--

Assume that the compiler allocates all the variables in the stack. The single arguments of *foo* and *bar* can be passed either by-value or by-reference. For each of the four possible combinations (i.e. both arguments by-value, *foo*'s argument by-value & *bar*'s argument by-reference, etc.) describe:

- What kind of data should the compiler put in the argument slots of the activation records for the calls on lines 6 and 11?
- What kind of assembly code will be necessary to retrieve the value of variable *a* on line 2?

$$[6 + (3 + 3) = 12]$$

7. (a) Consider the following basic block, in which all variables are integers and ****** denotes exponentiation:

```
a = b + c
z = a ** 2
x = 0 * b
y = b + c
w = y * y
u = x + 3
v = u + w
```

Assume that the only variables that are live at the exit of this block are *v* and *z*. In order, apply the following optimizations to this basic block. Show the result of each transformation.

- algebraic simplification
- common sub-expression elimination
- copy propagation
- constant folding

- (b) Discuss how induction variables can be detected and eliminated from the given intermediate code.

```
B2: j:= j+1
    t1:=4*j
    t2:=a[t1]
    if t2<10 goto B2
```

$$[(2 + 2 + 2 + 2) + 4 = 12]$$

INDIAN INSTITUTE OF ENGINEERING, SCIENCE AND TECHNOLOGY, SHIBPUR
 B.Tech 7th Semester Examinations, 2017
 Under 5-year Dual-Degree (B. Tech-M. Tech) Programme
 Compiler Design (CS 701)

Time: 3 hours

Full Marks: 70

(Answer question no - 1 and any five from the remaining)

1. (a) What is intermediate code? Why is it necessary for Compiler Design?
- (b) What are the characteristics of peephole optimization?
- (c) What are the rules to determine the leaders of basic blocks?
- (d) What is register descriptor and address descriptor?
- (e) List the advantages and disadvantages of an operator precedence parser.

[2 + 2 + 2 + 2 + 2 = 10]

2. Consider the following grammar G for Boolean expressions:

$$\begin{aligned} B &\rightarrow B \text{ or } T \mid T \\ T &\rightarrow T \text{ and } F \mid F \\ F &\rightarrow \text{not } B \mid (B) \mid \text{true} \mid \text{false} \end{aligned}$$

where B is the start symbol, set of non-terminals are $\{B, T, F\}$ and set of terminals are $\{\text{and, or, not, (,), true, false}\}$.

- (a) Compute *FIRST* and *FOLLOW* for each nonterminal in G .
- (b) Construct a predictive parsing table for G .
- (c) Show how your predictive parser processes the input string:

true and not false or true

Draw the parse tree traced out by your parser.

[6 + 3 + 3 = 12]

3. (a) Describe the relationship between a production and an item in an LR(0) grammar.
- (b) Consider the following grammar and construct the set of LR(0) items. Then fill out an SLR parse table for this grammar and indicate whether the grammar is ambiguous.

$$\begin{aligned} S &\rightarrow Aa \mid bAc \mid Bc \mid bBa \\ A &\rightarrow d \\ B &\rightarrow d \end{aligned}$$

Here S is the start symbol.

- (c) Explain *Phrase Level Error Recovery* strategy in LALR parsing method using suitable example.

[2 + 6 + 4 = 12]

4. (a) Give the *Syntax Directed Definition* to process sample statements like declarations in a procedure in C.
- (b) Explain how the types and relative addresses of declared names are computed and how scope information is dealt with.

- (c) How backpatching can be used to generate intermediate code for Boolean expressions and flow of control statements.

$$[3 + 6 + 3 = 12]$$

5. (a) Translate the expression $-(a + b) * (c + d) + (a + b + c)$ into: (i) quadruples, (ii) triples and (iii) indirect triples.
- (b) List the fields in an activation record. Write down the purpose of each of these fields in an activation record.
- (c) Explain the sequence of stack allocation process for a function call using a suitable example.

$$[3 + 5 + 4 = 12]$$

6. (a) What are the main operations for a symbol table? Discuss the data structures associated with a symbol table maintained as a list of hash tables and how the operations of a symbol table are implemented in that case. Give an example of what your symbol table would look like for a sample program.
- (b) How would you type check a for statement from C++?

$$[9 + 3 = 12]$$

7. (a) Consider the following sequence of three-address code:

```
x = 0
i = 0
L: t1 = i * 4
   t2 = a[t1]
   t3 = i * 4
   t4 = b[t3]
   t5 = t2 * t4
   x = x + t5
   i = i + 1
   if i < n goto L
```

- i. Draw a flow graph for this three-address code.
- ii. Optimize this code by eliminating common subexpressions, performing reduction in strength on induction variables, and eliminating all the induction variables that you can. State what transformations you are using at each optimization step.
- (b) Design a code generator algorithm and explain it with an example.

$$[(3 + 6) + 3 = 12]$$

INDIAN INSTITUTE OF ENGINEERING, SCIENCE AND TECHNOLOGY, SHIBPUR
B.E. 6th Semester (CST) Final Examination, April 2016
Compiler Design (CS 603)

Time: 3 hours

Full Marks: 70

(Answer Question no - 1 and any five from the rest)

1. (a) How to handle the *reserved* words and *identifiers* during recognition of tokens?
- (b) Why is it necessary to generate intermediate code instead of generating target program directly?
- (c) Explain the tasks of caller and callee when procedure is called and exit.
- (d) How do register allocation and evaluation order play an important role in a target code generation?
- (e) Explain the following with examples : (i) Quadruples (ii) Triples.

[2 × 5 = 10]

2. Consider the following Grammar

$$\begin{aligned} E &\rightarrow 5 + T \mid 3 - T \\ T &\rightarrow V \mid V * V \mid V + V \\ V &\rightarrow a \mid b \end{aligned}$$

where E is the start symbol, set of non-terminals are {E, T, V} and set of terminals are {5, 3, a, b}

- (a) Do the left factoring for the above grammar and compute the *FIRST* and *FOLLOW* set.
- (b) Construct a Predictive parsing table and check if the grammar is LL(1).
- (c) Show the parsing for the sentence, $5 + a * b$.

[6 + 3 + 3 = 12]

3. Consider the following grammar

$$\begin{aligned} S &\rightarrow CC \\ C &\rightarrow cC \mid d \end{aligned}$$

where S is the start symbol, set of terminals are {c, d}

- (a) Construct the sets of LR(1) items.
- (b) Construct the LALR parsing table for this grammar.
- (c) Show the parsing actions using the parsing table generated in (b) on the input "cdd".

[5 + 5 + 2 = 12]

4. (a) Give the syntax directed definition to process a sample variable declaration in C and construct dependency graph for the input *float x, y, z.*
- (b) List the fields in an activation record. Write down the purpose of each of these fields in an activation record. Explain the sequence of stack allocation process for a function call using a suitable example.
- (c) Describe the method of generating intermediate code for the flow-of-control statements.

[3 + 5 + 4 = 12]

5. (a) Write down the algorithm to find the leader in basic block. Write down the three-address code and construct the basic blocks for the following program segment.

```

sum = 0; i=0;
while (i <= 10)
{
    sum = sum + a[i];
    i++;
}

```

- (b) Why symbol-table is needed in various phases of compilers? How hashing can be used to design symbol-table.
- (c) Explain the characteristics of peephole code optimization technique.

[5 + 4 + 3 = 12]

6. (a) Consider the following code fragment and generate the equivalent three address code, compute the basic blocks, control flow graph and eliminate global common subexpression.

```

i = m-1;    j = n;
v = a[n];
while (1)
{
    do i = i+1; while (a[i] < v);
    do j = j-1; while (a[j] > v);
    if (i >= j) break;
    x = a[i]; a[i] = a[j]; a[j] = x;
}
x = a[i]; a[i] = a[n]; a[n] = x;

```

- (b) Explain different loop optimization techniques in detail on the control flow graph generated in (a).
- (c) How is the liveness of a variable calculated?

[6 + 4 + 2 = 12]

7. (a) Explain the role of declaration statements in the intermediate code generation with example.
- (b) Discuss the issues in the designing of a code generator.
- (c) Design a code generator algorithm and explain it with an example.

[3 + 6 + 3 = 12]

(CONFIDENTIAL)

BE 6th Semester Examination, 2015

Subject : Compiler Design

Paper/Code No CS 603

Branch: CST

Full Marks: 70

Answer Q. no. 1 and any four from the remaining questions.

1.

(a) Define synthesized attribute and inherited attribute associated with a grammar symbol. Show with an example the existence of an inherited attribute and a synthesized attribute for the grammar symbols present within a set of grammar rules. [3+3]

(b) What are the advantages and disadvantages associated with SLR(1), Canonical LR(1), and LALR(1) parsers. [3]

(c) Define leftmost canonical and rightmost canonical derivations. Show with an example that top-down parsing method is associated with leftmost canonical derivations whereas bottom-up parsing method is associated with reverse of rightmost canonical derivations. [2+3]

2.(a) Consider the following code fragment. Generate the three-address code for it. [2]

```
while c>d do
    {x= x+ y;
    if a < b then a = a + b; else c = c + d ;
    }
```

(b) Write the syntax directed translation schemes for the grammar rules if else, while and boolean expressions. [3 + 3 + 6]

3. (a) Given the grammar rules, compute FIRST and FOLLOW for all non-terminal symbols of the grammar consisting of the following production rules where S is the start symbol, {S,B,C,D,E,F} is the set of non-terminal symbols and {a,b,c,g,f,h} is the set of terminal symbols. [3+3]

$S \rightarrow aBDh$
 $B \rightarrow cC$
 $C \rightarrow bC \mid \epsilon$
 $D \rightarrow EF$
 $E \rightarrow g \mid \epsilon$
 $F \rightarrow f \mid \epsilon$

(b) Define LL(1) grammar. [2]

(d) Construct an LL(1) parsing table for the grammar and by looking into the entries of the table state whether the grammar is LL(1) or not. [5+1]

4.(a) State the conditions to be fulfilled for common sub-expression elimination and loop optimization. [3]

(b) Explain with example how DAG can be used for common sub-expression elimination. [5]

(c) Define available expression data flow property. Which purpose is this data flow property useful for? State data flow equations for available expression data flow property.

[2+1+3]

5. (a) What are the criteria required to be satisfied for static storage allocation. [2]

(b) Define activation record. What are the different parts of activation record? Define activation base pointer. Why is previous value of activation base pointer stored in activation record during execution of a block-structured program segment? Explain with example. [1+1 + 1 + 2+2]

(c) What is the role of display? How will display be created when a function at level j ($j < i$) will be entered from a function at level i ? [1+4]

6. (a) Define Pass. Why is design of a two-pass assembler required? What are the data structures needed for designing an assembler? Draw the flowchart for assembler design in Pass 1? How will you ensure design of a single-pass assembler? [1+1+2+4+2]

(b) What is the objective behind using MACRO in an assembly language program? Draw the flowchart for designing MACRO Assembler, defining the necessary data structures.

[1+3]

7. Given the grammar with S being the start symbol, $\{S, A, B, C\}$ being the set of non-terminal symbols and $\{\#, a, b, c, d, e\}$ being the set of terminal symbols and the rules are as follows:

0. $S \rightarrow A\#$

1. $A \rightarrow bB$

2. $B \rightarrow cC$

3. $B \rightarrow cCe$

4. $C \rightarrow dA$

5. $A \rightarrow a$

(a) Generate the sets of LR(1) items.

[10]

(b) Is the grammar LR(1)? If not, why not?

[4]

15.5.14
L

BE 6th semester Examination 2014
Subject : Compiler Design Code: CS 603
Full Marks : 70 Time: 3 hours

Answer question no. 1 and any four from the remaining questions.

1.(a) Justify the statement:

‘ Interpreter is more suitable than compiler in development environment whereas the reverse is true in production environment.’ [2]

(b) Define leftmost canonical derivation and rightmost canonical derivation. Show with an example that top-down parsing method is associated with leftmost canonical derivations whereas bottom-up parsing method is associated with reverse of rightmost canonical derivations. [2+3]

(c) Distinguish among SLR(1), LR(1) and LALR(1) parsers with reference to number of states, economy and efficiency. [3]

(d) Why is design of a multi-pass compiler needed instead of a single-pass one? [2]

(e) Justify the presence of a look-ahead buffer within a lexical analyzer. [2]

2.(a) What are the basic features of assembly language statement? What are the advantages of assembly language compared to the machine language? Describe the method for designing a single-pass assembler using a suitable data structure. [2+2+4]

(b) What is a MACRO? What are the data structures needed for designing a macro preprocessor. Draw the flowchart for a macro pre-processor. [1+1+4]

3. (a) Differentiate between static storage allocation and dynamic storage allocation. What are the criteria that need to be satisfied for static storage allocation? Define activation record. What are the roles of activation base pointer and display in activation record? How the display will be constructed when a block at level j is entered from a block at level i ? [2+2+1+2+3]

(b) Define basic block. Explain with example how DAG can be used for variable propagation and common sub-expression elimination. [1+ 3]

4. Given the grammar with the set of terminal symbols $\{\#, =, +, *\}$, set of non-terminal symbols $\{S, G, E, T\}$ and the start symbol S , where production rules are as follows:

$S \rightarrow G\#$

$G \rightarrow E = E \mid f$

$E \rightarrow T \mid E + T$

$T \rightarrow f \mid T * f$

(a) Compute FIRST and FOLLOW for all non-terminal symbols excluding S of the grammar. [3]

(b) Construct LR(0) machine for this grammar. [5]

(c) Mark all ambiguous states of the grammar. Show their item sets and describe which type of conflict is associated with each of those states. [4]

(d) Is the grammar SLR(1)? [2]

5. Define LL(1) grammar.

For the given grammar with set of terminal symbols $\{a, b, c\}$, set of non-terminal symbols $\{S, A, B, C, D\}$ and start symbol S , find FIRST and FOLLOW for all non-terminal symbols, construct the parsing table and tell whether the following grammar is LL(1) or not.

$S \rightarrow aABC$

$A \rightarrow a$

$A \rightarrow bbD$

$B \rightarrow a$

$B \rightarrow \epsilon$

$C \rightarrow b$

$C \rightarrow \epsilon$

$D \rightarrow c$

$D \rightarrow \epsilon$

[2+5+6+1]

6.(a) Consider the following code fragment. Generate the three-address code for it.

*if $a < b$ then while $c > d$
 $\{c = c + b\}$
 else $c = c + d$;*

[2]

(b) Write the syntax directed translation schemes for the grammar rules involving boolean expressions, if else and while statement and generate annotated parse tree for the code fragment.

[5 + 2 + 2 + 3]

7.(a) Define the following operator precedence relations

[3]

$<, >, =$

(b) Define operator precedence grammar.

[1]

(c) Compute Leading and Trailing of the non-terminal symbols for the grammar consisting of the following rules with the set of terminal symbols $\{+, -, *, /\}$

$E \rightarrow T \mid E+T \mid E-T$

$T \rightarrow F \mid T * F \mid T / F$

[4]

$F \rightarrow P \mid F \mid P$

$P \rightarrow i \mid (E)$

(d) Derive the operator precedence relations existing between pair of terminal symbols.

[6]

BE 7th Semester Examination, 2011**Subject : Compiler Design****Paper/Code No CS 703****Branch: CST****Full Marks: 70****Answer Q. no. 1 and any four from the remaining questions.**

1. (a) What are the advantages of a translator module? [2]
 (b) What are the reasons behind writing a multi-pass compiler rather than a single-pass one? [2]
 (c) What do you mean by an ambiguous grammar? Explain with the following grammar and parse tree, how ambiguity takes place for the string 3-2+1, where {expr, 'digit'} is the set of non-terminal symbols, expr is the start symbol of the grammar and {+, -, 0, 1, 2, ..., 9} is the set of terminal symbols of the grammar. [1+3]

expr $\rightarrow$ expr + expr | expr - expr | digit
 digit $\rightarrow$ 0|1|2|3|4|5|6|7|8|9

- (d) Define four types of grammar. [4]
 (e) Inserting a code-optimizer within a compiler increases compilation time. How would you justify the presence of code-optimizer within a compiler? [2]

2. (a) Construct the LR(0) finite control transition diagram for the grammar with the set of terminal symbols $V_T = \{x, y, z, a, q, \#\}$, set of non-terminal symbols $V_N = \{S', S, A, B\}$, # being a special terminal symbol used as end-marker, with the set of production rules as follows: [5]

$S' \rightarrow S \#$
 $S \rightarrow xAy \mid xBy \mid xAz$
 $A \rightarrow aS \mid q$
 $B \rightarrow q$

- (b) Compute FIRST and FOLLOW of all non-terminal symbols except S' . Construct the LR parsing table and from the table tell whether the grammar is SLR or not. [(2+3)+4]

3. (a) Consider the following code fragment. Generate the three-address code for it. [2]
 while c > d do

{ x = x + y;
 if a < b then a = a + b; else c = c + d;
 }

- (b) Write the syntax directed translation schemes for the grammar rules if else, while and boolean expressions. [3 + 3 + 6]

4. (a) State the disadvantages of a top-down parser using brute-force approach. [3]
 (b) How can left-recursion be removed from a grammar? [2]
 (c) Define $FIRST(\alpha)$ and $FOLLOW(A)$, $\alpha \in V^+$ and $A \in V_N$, where V is the alphabet and V_N is the set of non-terminal symbols. [3]
 (d) Define LL(1) grammar. [2]
 (e) Discuss about recursive descent parser. How does left-factoring help design recursive descent parser? [3 + 1]

- 5.(a) State the conditions to be fulfilled for common sub-expression elimination and loop optimization. [3]
 (b) Explain with example how DAG can be used for common sub-expression Elimination. [5]
 (c) Define available expression data flow property. Which purpose is this data flow property useful for? State data flow equations for available expression data flow property. [2+1+3]

- 6.(a) Define synthesized attribute and inherited attribute. What is an L-attributed definition. [2+2+2]

Explain how L-attributed definition will be implemented during predictive parsing while evaluating the expression $9-5+2$ using the following grammar rules when $\{ +, -, (,), \text{num} \}$ is the set of terminal symbols and $\{ E, R, T \}$ is the set of non-terminal symbols and E is the start symbol of the grammar. [8]

$E \rightarrow T R$
 $R \rightarrow + T R \mid - T R$
 $R \rightarrow \epsilon$
 $T \rightarrow (E)$
 $T \rightarrow \text{num}$

7. (a) What are the criteria required to be satisfied for static storage allocation. [2]
 (b) Define activation record. What are the different parts of activation record?
 Explain the role of static link and display in activation record during execution by using a suitable program segment. [2 + 1 + (2+2)]
 (c) How will display be created when a function at level j ($j < i$) will be entered from a function at level i ? [5]